

COSC2670/2738 Assignment 3

Practical Data Science with Python
COSC 2670/2738
Assignment 3

Assessment Type: Individual Marks: 30

Due Date: 23:59 on the 14th of June, 2026

Please read this carefully before attempting

This is an individual assignment. You are not allowed to collude with any other people or plagiarise their work. You are expected to present the results of your own thinking and writing. Never copy other student's work (even if they "explain it to you first") and never give your written work to others. Keep any conversation high-level and never show your solution to others. Never copy from the Web or any other resource. Remember you are meant to generate the solution to the questions by yourself. Suspected collusion or plagiarism will be dealt with according to RMIT policy.

In the submission you will be required to certify that the submitted solution represents your own work only by agreeing to the following statement:

I certify that this is all my own original work. If I took any parts from elsewhere, then they were non-essential parts of the assignment, and they are clearly attributed in our submission. I will show I agree to this honor code by typing "Yes":
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Introduction

This assignment focuses on recommender systems, especially item-based collaborative filtering. You are required to reproduce the baseline method from the classic paper by Sarwar et al. (2001), and then conduct two controlled improvement experiments.

Here is the paper Sarwar et al. (2001), and you can also find it from the assignment canvas page.

- Sarwar, B., Karypis, G., Konstan, J., & Riedl, J. (2001). Item-Based Collaborative Filtering Recommendation Algorithms.

Dataset

You must use the MovieLens 100K dataset provided in the course materials. The dataset contains user ratings for movies. You should use the rating file to construct a user-item rating matrix.

You may use Python libraries such as pandas, numpy, matplotlib, and sklearn for data preparation, numerical calculation, visualisation, splitting, and evaluation.

General Requirements

You need to use the provided *framework.ipynb* file to complete your assignment. This Python framework file, named *assignment3_framework.ipynb*, is designed to help you get started and will also automate the correctness marking process. The framework includes both the training data and the test data.

Please read the comments in the provided assignment framework carefully and insert your own code only in the designated code cells as indicated.

Please do not change anything else in the remaining cells of the framework, as doing so may result in errors during the automatic marking process, which might cause the submission invalid. Please provide detailed comments to explain your implementation.

- Your notebook must be named *Assignment3.ipynb*.
- Before your submission, your notebook must run from start to finish using Kernel -> Restart & Run All.
- You must remove unnecessary code cells, failed attempts, debugging outputs, and unrelated files.
- You must clearly cite any sources you use, including research papers, online resources, and documentation.
- You must not hard-code final results. Results should be generated by your code.
- All experiments must use only the training data to build models. Test data must only be used for evaluation.
- AI tools are not allowed to be used as generative tools in this assignment.

Tasks

Task 1: Reproduce the Sarwar Item-Based CF Baseline Model (10)

In this task, you are required to read Sarwar et al. (2001), *Item-Based Collaborative Filtering Recommendation Algorithms*, and reproduce the main item-based collaborative filtering method described in the paper using the MovieLens 100K dataset. Your implementation should apply the paper's item-based collaborative filtering idea to the provided MovieLens data, with the following specific requirements:

- Similarity computation: You must use **adjusted cosine similarity** to compute item-item similarities, as described in Section 3.1.3 of the paper.
- Prediction computation: Instead of the linear regression model described in Section 3.2.2 of the paper, you must use the KNN-regression approach taught in our course. Use the **k=20** most similar items as neighbours and apply the regression-based prediction method covered in class.

Task 2: Controlled Improvement Experiments (10)

In this task, you must conduct two improvement experiments based on your Task 1 baseline model. You must choose **one option from Group 1 (Option A or Option B)** and **one option from Group 2 (Option C or Option D)**.

You must compare each selected improvement with the Task 1 baseline model.

Group 1	Direction	Minimum Requirements
Option A	Similarity Measure Improvement	Replace the baseline adjusted cosine similarity with cosine similarity and Pearson correlation . Compare the

		results with the baseline.
Option B	Prediction Method Comparison	Replace the baseline KNN-regression with the weighted sum method described in Section 3.2.1 of the paper (Sarwar et al. (2001)). Compare MAE/RMSE with the baseline.

Group 2	Direction	Minimum Requirements
Option C	Neighbourhood Size Tuning	Test 5 different values of k , ($k = 5, 10, 20, 30$, and 50) and identify the best-performing k base on the baseline in Task1.
Option D	Significance Weighting	Apply significance weighting to reduce the influence of item pairs with very few co-rated users. For each pair of items (i, j), compute the number of users who rated both items, denoted as $c(i, j)$. Then apply the following weighting to the similarity matrix: $weighted_sim(i, j) = sim(i, j) \times min(c(i, j), T) / T$ You must test the following threshold values: $T = 10, 20, 30, 50$. Use $k = 20$ for all experiments. Report MAE and RMSE for each threshold value alongside the baseline (no weighting).

The experiments in each section are independent. Except for the specific part and parameter that need to be tested in the corresponding option, all other parameters should remain the same as the baseline settings. For example, in Option D, you need to test T based on the baseline, while still using adjusted cosine similarity, KNN-regression, and $K=20$.

Task 3: Presentation (10)

- The presentation should
 - Provide a brief summary of Sarwar et al. (2001), *Item-Based Collaborative Filtering Recommendation Algorithms*. Particularly, you need to answer: how user-item ratings are represented, how item-item relationships are measured using adjusted cosine similarity, how unknown ratings are predicted using KNN-regression, and how prediction performance is evaluated.
 - For Task 2, depending on the two options you selected, your presentation must include the corresponding analysis:
 - Option A: Compare the results with the baseline and discuss why adjusted cosine may or may not perform better.
 - Option B: Compare MAE/RMSE with the baseline and discuss the advantages and disadvantages of each prediction approach.
 - Option C: Present results in a table or graph, identify the best-performing k , and analyse the result.
 - Option D: Compare the results with the baseline and explain why significance weighting does or does not improve performance.

- The presentation should be no more than 8 minutes.
- Your presentation slides should be:
 - Microsoft PowerPoint slides (with audio inserted for each slide by using: **Insert** → **Audio** → **Record Audio**).
 - or you can create your own presentation slides (e.g. PDF version) and please submit your own recording (in the format of mp4 or avi) of your presentation as well.

Submission Requirements

The assignment is due at **23:59 on the 14th of June, 2026**. Assignments submitted after this time will be subject to standard late submission penalties.

Submit the following files:

- Assignment3.ipynb
- Slides.pdf and your presentation recording in the required format, or Your Microsoft PowerPoint slides (with audio inserted for each slide).

Before submission, you must:

- Restart the notebook kernel.
- Run all cells from beginning to end.
- Check that all outputs are displayed correctly.
- Remove unnecessary cells and temporary files.

All files must be submitted as one single zip file named with your student number, for example 1234567.zip. Please do not submit unrelated or unnecessary files.

Marking Summary

Task	Marks
Task 1: Sarwar baseline item-based CF reproduction	10
Task 2: Two controlled improvement experiments	10
Task 3: Presentation	10
Total	30

